

Algebra I

Review for Midterm Examination Winter 2025

Name: _____

Please show all work clearly for full credit.

The midterm exam is a cumulative exam that will be composed of

55 multiple-choice questions covering all material from Units 1 – 4.

A graphing calculator will be permitted for all questions.

Tutorials:	Monday, December 15 th	2:30 – 4:00 p.m.
Exam:	Tuesday, December 16th	8:00 – 10:15 a.m.

Materials to bring:

- *pencils and eraser
- *a graphing calculator
- *This review, completed
- *3x5 notecard
- (handwritten, front and back)

For credit, **completed work (not just answers)** must be shown neatly.

Turn in at final exam for a 40-point completion grade.

2025-26 Algebra I Midterm Study Guide

Unit 1

1. Use the power rule to simplify each expression:

a. $(2x^2y^2)^3$

b. $(-3xy)^2$

2. Use the quotient rule to simplify each expression:

a. $\frac{4s^5p^9}{-2sp^3}$

b. $\frac{-12x^8y^7}{6x^2y^4}$

3. Simplify each expression

a. $(-2x^4y^5 \cdot xy^3)^3$

b. $\left(\frac{4x^2y^3}{z^7}\right)^3$

4. Simplify each expression

a. $\sqrt{18}$

b. $-2\sqrt{18}$

c. $-5\sqrt{32}$

d. $\sqrt{300x^5y^6}$

e. $\sqrt{500x^3y^8}$

f. $\sqrt{\frac{64}{25}}$

g. $\sqrt{45} + \sqrt{20}$

h. $3\sqrt{18} + 2\sqrt{8}$

i. $\sqrt{3} \cdot \sqrt{5}$

j. $\sqrt{7} \cdot \sqrt{2}$

Unit 2

5. Evaluate each expression according to the order of operations.

a. $2 + 30 \div 6 \cdot 3 - 20$

b. $10 \div 2(3^2 + 15 \div 2 \cdot 4)$

6. Represent each statement as an algebraic expression:

a. Twice the sum of a number and eleven

b. Half the difference of a number and six

c. Two less than five times the sum of x and y

d. Two more than four times the difference of x and y

7. Solve each equation:

a. $16 + \frac{5}{8}x = 56$

b. $7(x - 2) - 6x = 6x + 5 - 5(x + 3)$

c. $-\frac{1}{9}y + \frac{5}{2} = \frac{1}{2}$

d. $5(x + 1) - x = 6x - 2(x - 3) - 1$

e. $6 = \frac{1}{4}(8x + 4)$

f. $8x - 3(3x + 8) = -6(-6 + x)$

g. $\frac{3}{2}x - \frac{1}{4} = \frac{1}{2}$

h. $-2(x + 3) + 5x = -(-2x - 1) + 3$

8. Solve the literal equation for the indicated variable:

a. $px + 8 = M$; (for x)

b. $3n + 3t = 15$; (for t)

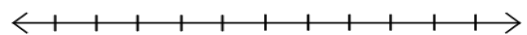
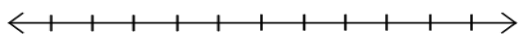
c. $t - w = pa$; (for a)

d. $P = 2W + 2L$; (for L)

9. Solve each inequality and graph the solution on the number line. Also, write your answer in interval notation.

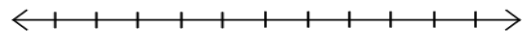
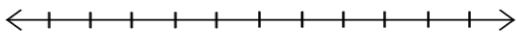
a. $-3x + 7 < 4$

b. $7x + 32 \leq 67$



c. $-5(3 - 2x) - 59 \geq 3(x + 1)$

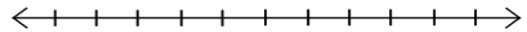
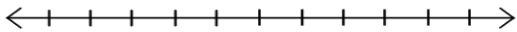
d. $3x - 5 < 8x + 10$



10. Solve each inequality and graph the solution on the number line.

a. $2x + 2 > 6$ or $-2x - 1 \geq 7$

b. $3 + 3x \geq -6$ and $4x - 2 < 10$



11. Quincy's age is 4 years less than half her aunt's age, x . If Quincy is 14 years old, write an equation that can be used to determine her mother's age.

12. The length of a rectangle is 8 cm less than 4 times the width. If the perimeter is 240 cm, find the width and the length of this rectangle.

13. Sammy's weekly wages are \$30 less than twice what Noa's earns in a week, x . If Sammy makes \$150 a week, write an equation that can be used to determine Noa's weekly wages.

Unit 3

14. Which one of the following tables represents a function?

A.

x	3	5	7	9
f(x)	2	4	2	4

B.

x	2	4	2	4
f(x)	3	5	7	9

C.

x	0	1	-1	0
f(x)	0	-1	0	1

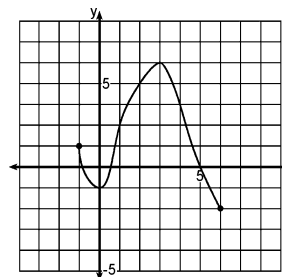
D.

x	0	-1	0	1
f(x)	0	1	-1	0

15. What is the **domain and range, in interval notation**, of the function shown in the graph below?

Domain: _____

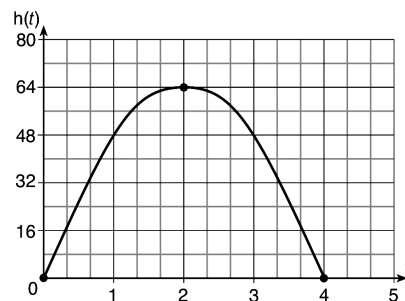
Range: _____



16. The diagram below shows the graph of h , which models the height, in feet, of rocket t seconds after it was shot into the air. What is the domain and **range** of h ?

Domain: _____

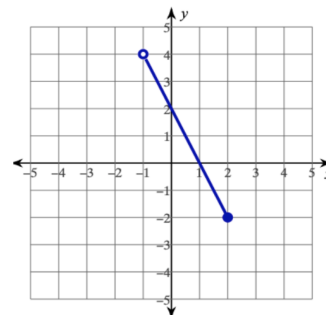
Range: _____



17. A function, f , is represented by the line segment drawn in the accompanying graph. What is the **domain and range** of function f ?

Domain: _____

Range: _____



18. The table below defines the function f . According to the function shown, what is the value of each of the following?

a. $f(-1) =$ _____

b. $f(1) + f(2) =$ _____

c. For what value of x is $f(x) = 1$?

d. For what value of x is $f(x) = -3$?

x	f(x)
-1	5
0	1
1	-3
2	-7

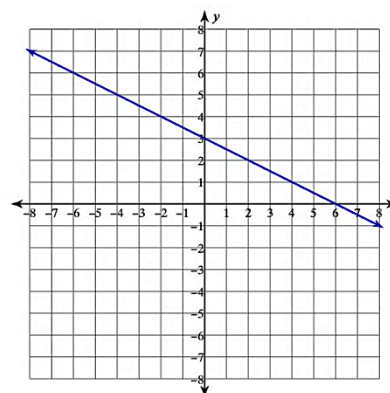
19. The linear function f is defined by the graph below. Determine the following:

a. $f(4) =$ _____

b. $f(0) =$ _____

c. For what value of x is $f(x) = 4$?

d. For what value of x is $f(x) = 1$?



20. If $f(x) = -2x - 3$, determine:

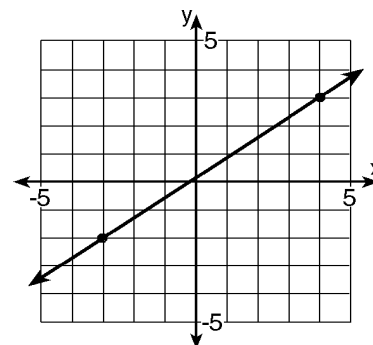
a. $f(-2) =$ _____

b. $f(0) =$ _____

c. For what value of x is $f(x) = 5$?

d. For what value of x is $f(x) = -1$?

21. Using the accompanying graph, determine the rate of change of y with respect to x .



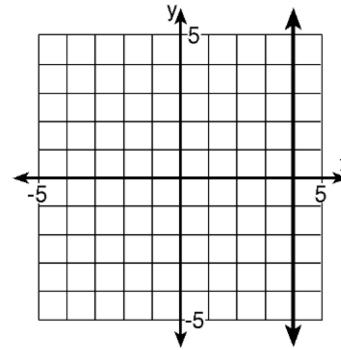
22. Find the equation of each linear function represented by the tables below in slope-intercept form.

x	0	1	2	3
y	9	6	3	0

23. Given the points $(6, 4)$ and $(7, 2)$, what is the slope of the line passing through these points?

Unit 4

24. Write the equation and slope of the line in this graph?



25. What is the slope of the line whose equation is $3x - 4y = 16$?

26. What is the slope of the line whose equation is $7x + 5y = 15$?

27. Rewrite the equation $4x + 2y = 8$ in slope intercept form.

28. Rewrite following equations that is equivalent to in **slope-intercept** form:

$$y - 2 = -\frac{1}{3}(x - 9)$$

29. The equation of a line is $x = 0$. Which is a description of this line?

30. The equation of a line is $y = 0$. Which is a description of this line?

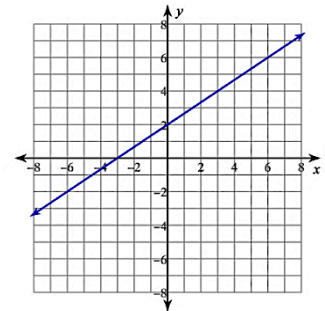
31. Using the **slope-intercept form**, what is the equation of the line that passes through the point $(4, -2)$ with a slope of -3 ?

32. Using the **slope-intercept form**, what is the equation of the line that passes through the point $(-3, 5)$ with a slope of $-\frac{1}{3}$?

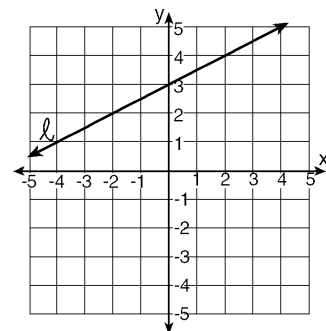
33. What is the **slope-intercept** equation of line through the points $(10, 4)$ and $(0, 2)$?

34. What is the **slope-intercept** equation of line through the points $(-4, 4)$ and $(0, -1)$?

35. Write the equation of the line shown in standard form.



36. What equation represents line ℓ , shown in the accompanying diagram?



37. What are the x- and y-intercepts of the line with an equation $5x + 2y = 10$?

x-intercept: _____

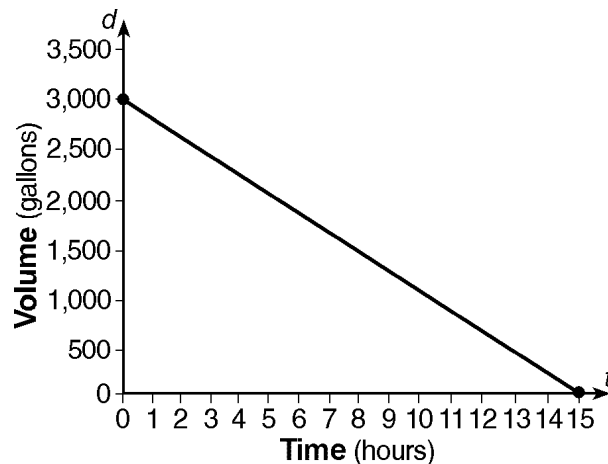
y-intercept: _____

38. What are the x- and y-intercepts of the line with an equation $x - 6y = 18$?

x-intercept: _____

y-intercept: _____

39. A hole in a water storage tank causes the tank to completely drain. The graph below shows the relationship between the amount of water left in the tank and the time that has elapsed.



Which one of the following statements accurately describes the rate at which the water is flowing from the tank?

- A. The water is flowing out of the tank at a rate of 200 gallons per hour.
- B. The water is flowing out of the tank at a rate of 20 gallons per minute.
- C. The water is flowing out of the tank at a rate of 20 gallons per hour.
- D. The water is flowing out of the tank at a rate of 200 gallons per minute.

40. The cost of airing a commercial on television is modeled by the function

$$C(n) = 110n + 900,$$

where n is the number of times the commercial is aired.

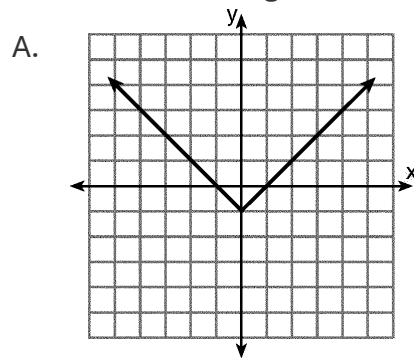
Based on this model, which statement is true?

- A. The commercial costs \$110 to produce and \$900 each time it is aired.
- B. The commercial costs \$1,010 to produce and can air an unlimited number of times.
- C. The commercial costs \$900 to produce and \$110 each time it is aired.
- D. The commercial costs \$0 to produce and \$110 per airing up to \$900.

41. The phone company charges a standard fee of \$24.95 per month plus \$0.07 per minute for long distance calls. What is an equation that can be used to represent the monthly cost (c) if a family makes long distance phone calls totaling d minutes?

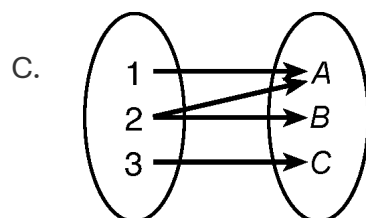
- A. $c = 24.95 + 0.07d$
- B. $c = 24.95d$
- C. $c = 24.95d + 0.07$
- D. $c = 0.07d$

42. Which one of the following relations is **not** a function?



B.

x	y
-10	-2
-6	2
-2	6
1	9
5	13



D. $\{(2, 7), (3, 9), (4, 11)\}$

Answer Key

Unit 1: Exponents and Radicals

1. a. $8x^6y^6$ b. $9x^2y^2$
 2. a. $-2s^4p^6$ b. $-2x^6y^3$
 3. a. $-8x^{15}y^{24}$ b. $\frac{64x^6y^9}{z^{21}}$
 4. a. $3\sqrt{2}$ b. $-6\sqrt{2}$ c. $-20\sqrt{2}$ d. $10x^2y^3\sqrt{3x}$ e. $10xy^4\sqrt{5x}$ f. $\frac{8}{5}$ g. $5\sqrt{5}$ h. $13\sqrt{2}$ i. $\sqrt{15}$
j. $\sqrt{14}$
-

Unit 2: Solving Equations and Inequalities

5. a. -3 b. 195
 6. a. $2(n + 11)$ b. $\frac{1}{2}(n - 6)$ c. $5(x + y) - 2$ d. $4(x - y) + 2$
 7. a. $x = 64$ b. No Solution ($\emptyset$) c. $y = 18$ d. All Real Numbers ($\mathbb{R}$) e. $x = \frac{5}{2}$ f. $x = 12$ g.
 $x = \frac{1}{2}$ h. $x = 10$
 8. a. $x = \frac{M-8}{p}$ b. $t = 5 - n$ c. $a = \frac{t-w}{p}$ d. $L = \frac{P}{2} - W$
 9. a. $x > 1$; Interval: $(1, \infty)$ b. $x \leq 5$; Interval: $(-\infty, 5]$ c. $x \geq 11$; Interval: $[11, \infty)$ d.
 $x > -3$; Interval: $(-3, \infty)$
 10. a. $x > 2$ or $x \leq -4$ b. $-3 \leq x < 3$
 11. $14 = \frac{1}{2}x - 4$
 12. Width: 25.6 cm; Length: 94.4 cm
 13. $150 = 2x - 30$
-

Unit 3: Functions and Relations

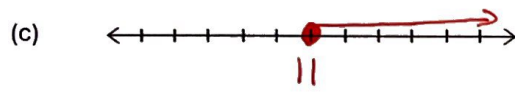
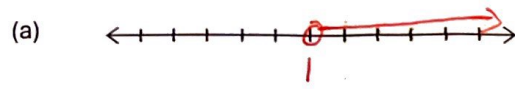
14. A
15. Domain: $[-1, 6]$; Range: $[-2, 6]$
16. Domain: $[0, 4]$; Range: $[0, 64]$
17. Domain: $(-1, 2]$; Range: $[-2, 4)$

18. a. $f(-1) = 5$ b. $f(1) + f(2) = -10$ c. $x = 0$ d. $x = 1$
19. a. $f(4) = 1$ b. $f(0) = 3$ c. $x = -2$ d. $x = 4$
20. a. $f(-2) = 1$ b. $f(0) = -3$ c. $x = -4$ d. $x = -1$
21. $m = \frac{5}{7}$
22. $y = -3x + 9$
23. $m = -2$
-

Unit 4: Linear Functions

24. Equation: $x = 4$; Slope: Undefined
25. $m = \frac{3}{4}$
26. $m = -\frac{7}{5}$
27. $y = -2x + 4$
28. $y = -\frac{1}{3}x + 5$
29. y-axis
30. x-axis
31. $y = -3x + 10$
32. $y = -\frac{1}{3}x + 4$
33. $y = \frac{1}{5}x + 2$
34. $y = -\frac{5}{4}x - 1$
35. $2x - 3y = 6$
36. $y = \frac{1}{2}x + 3$
37. x-intercept: 2; y-intercept: 5
38. x-intercept: 18; y-intercept: -3
39. A. The water is flowing out of the tank at a rate of 200 gallons per hour.
40. C. The commercial costs \$900 to produce and \$110 each time it is aired.
41. A. $c = 24.95 + 0.07d$
42. C. (The mapping diagram where a single input '2' has two outputs 'A' and 'B')

9.



10.

